Case Title

Geolocation of Accommodation and Café Locations from Social Media Images

Objective

The objective of this investigation was to determine the geographic locations of two images posted on social media using open-source intelligence (OSINT) techniques, specifically through reverse image analysis and visual comparison.

Case Background

Two images were posted by a subject during travel within Africa. The first image was associated with accommodation, with a partial indication that the location was in West Africa. The second image depicted a café environment at an unknown location. No metadata or geolocation tags were available for either image, requiring analysis based solely on visual and open-source information.

Case 1: Accommodation Location

Evidence Collected

- **Exhibit 1 – Subject Image (Accommodation)**

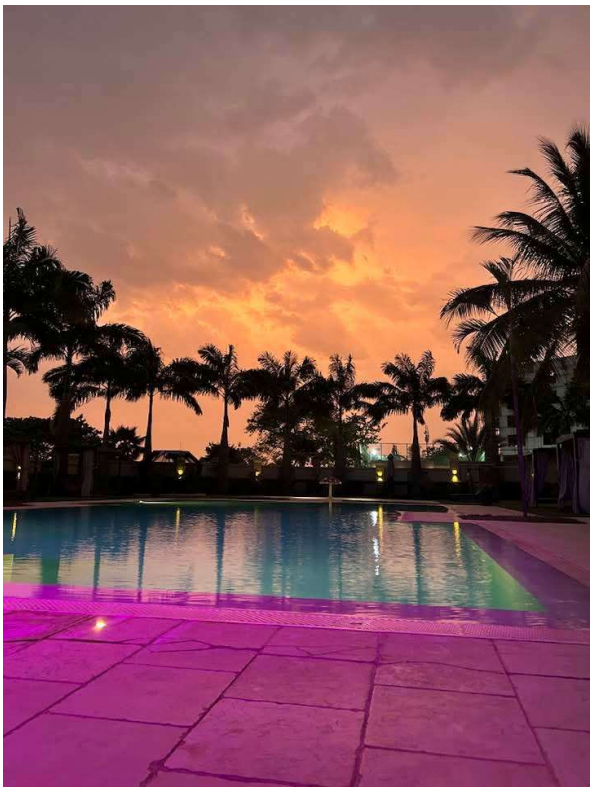


Exhibit 2 – Reference Image A

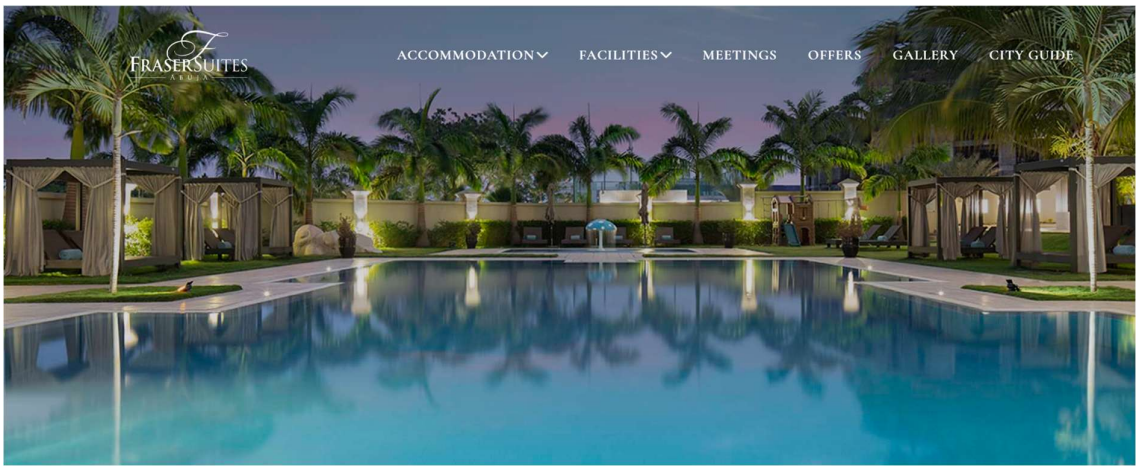


Exhibit 3 – Reference Image B

Methodology

A reverse image search was conducted using Google Reverse Image Search to identify visually similar images related to the subject image.

Search results were analyzed and filtered based on relevance to the known regional hint (West Africa).

Two independent reference images were selected based on strong visual similarity.

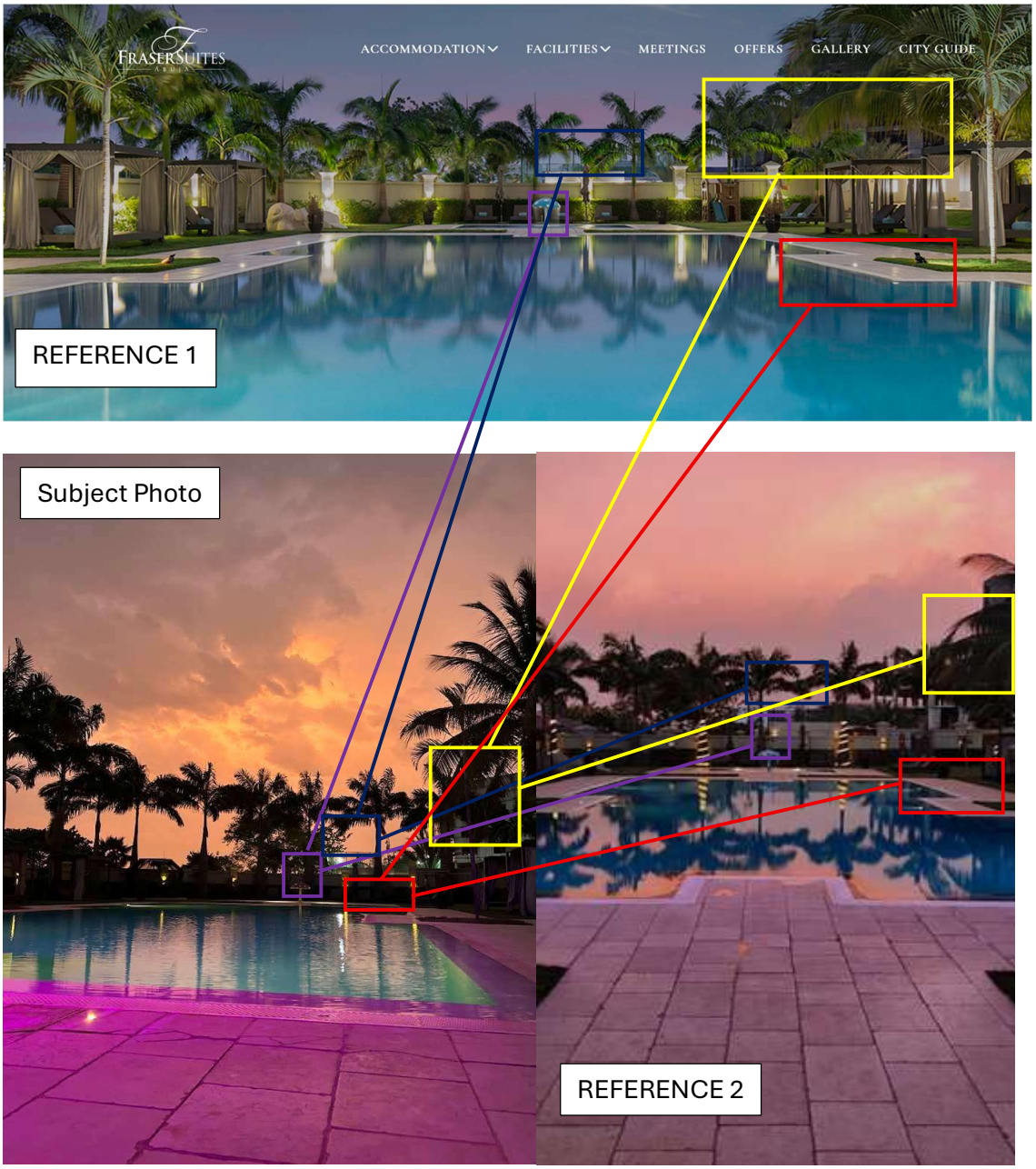
A comparative visual analysis was then conducted, focusing on fixed environmental and structural features such as vegetation patterns, building design, and spatial layout.

Analysis

The subject image was compared against the selected reference images, revealing several consistent features:

- The alignment and spacing of the tree line were consistent across all images
- The building structure, including layout and architectural design, matched closely
- The spatial relationship between the building and surrounding vegetation was identical
- Environmental features appeared stable and non-random, indicating a fixed location

These consistent elements across multiple independent sources strongly support that all images were taken at the same location.



Findings

The location of the accommodation was identified as:

- **City:** Abuja, Nigeria
- **Area:** Central Business District
- **Specific Location:** 294 Leventis Close, Ayya's The Fraser Suites Hotel

- **Confidence Level**
- **High**
- This conclusion is supported by multiple independent image matches and the presence of consistent environmental and structural features. The likelihood of these features aligning coincidentally in a different location is low.

Limitations

- This analysis is based solely on visual comparison and publicly available images. No metadata, GPS data, or insider information was available to independently verify the location.

Case 2: Café Location

Evidence Collected

Exhibit 4 – Subject Image (Café)

Exhibit 5 – Reference Image A

Exhibit 6 – Reference Image B

Methodology

A reverse image search was conducted using Google Reverse Image Search to identify images with similar visual characteristics to the subject image. Hint was Ethiopia.

Relevant results were identified and cross-referenced with publicly available sources.

Two independent reference images were selected based on strong visual similarity to the subject image.

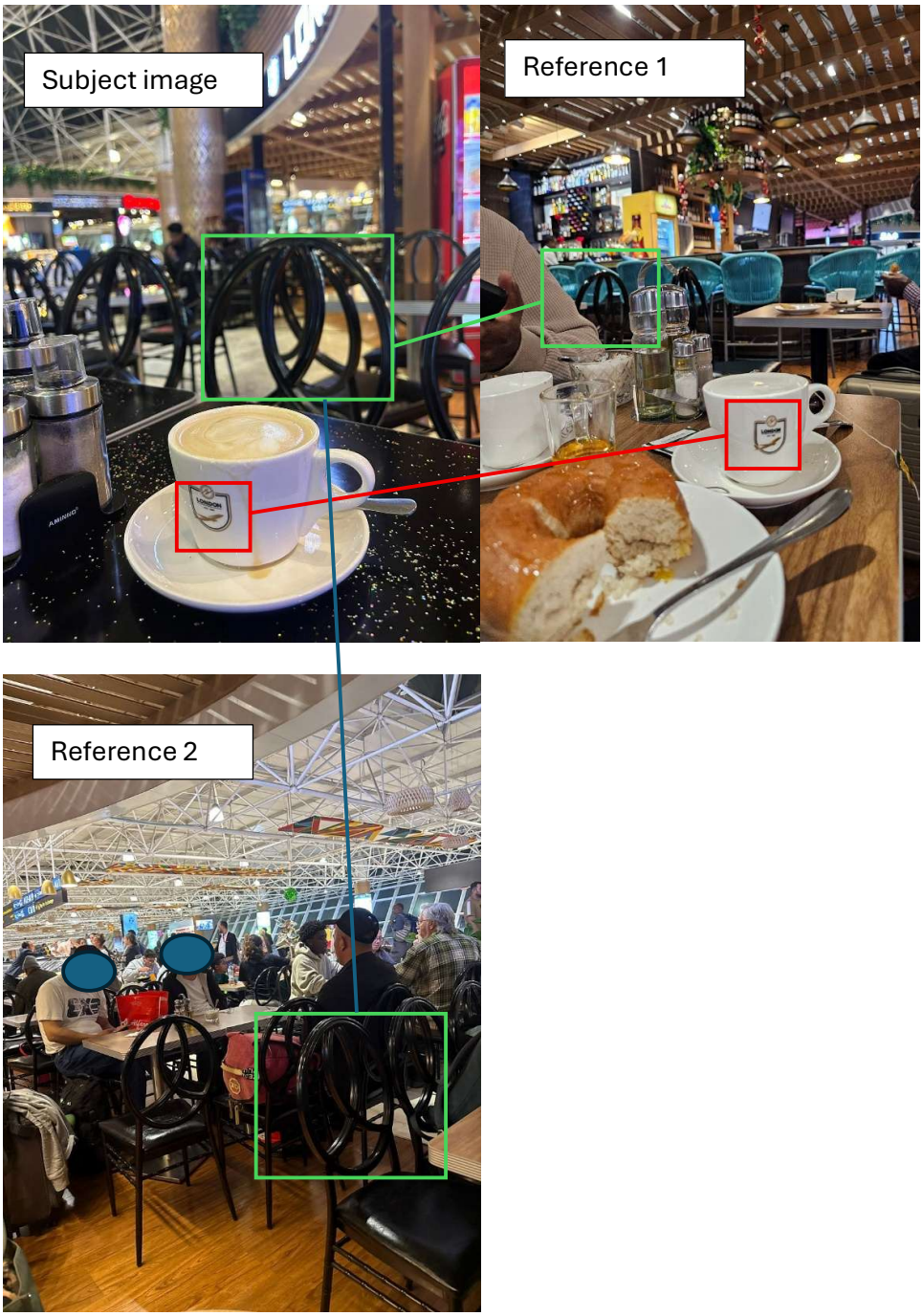
A detailed visual comparison was conducted focusing on architectural design, seating layout, and spatial arrangement.

Analysis

Comparison between the subject image and reference images revealed the following:

- The seating layout and arrangement were identical across images
- Architectural design and structural elements matched precisely
- The positioning and spacing of objects within the café were consistent
- The overall spatial configuration of the environment was unique and matched across all sources

These features collectively confirm that the images were taken at the same location.



Findings

The café location was identified as:

- **City:** Addis Ababa, Ethiopia
- **Venue:** LODON Café & Satellite Restaurant, Addis Airport

- **Confidence Level**
- **High**
- The conclusion is supported by direct visual matches across multiple independent sources and the presence of unique environmental and structural characteristics that are unlikely to be replicated elsewhere.

- **Limitations**
- The analysis relied exclusively on visual comparison and publicly available sources. No metadata or geolocation data was available to independently confirm the findings.

- **Analytical Summary**
- This investigation demonstrates the effectiveness of reverse image search combined with structured visual analysis in determining geographic locations. By validating findings across multiple independent sources and focusing on consistent environmental and architectural features, reliable conclusions were reached without reliance on metadata or privileged information.

Skills Demonstrated

- Image-based geolocation
- Open-source intelligence (OSINT) techniques
- Source validation and cross-referencing
- Visual pattern recognition
- Investigative documentation